

DOFBOT Dancing

1. Experiment Approach

This experiment controls the DOFBOT to dance by modifying the angles of different servos and adding delay times to create a dancing effect.

2. Code Content

Code path: /home/yahboom/Dofbot/3.ctrl_Arm/7.dance.ipynb

```
#!/usr/bin/env python3
#coding=utf-8
import time
from Arm_Lib import Arm_Device
# Create DOFBOT object
Arm = Arm_Device()
time.sleep(.1)
time_1 = 500
time_2 = 1000
time_sleep = 0.5
```

```
# DOFBOT loop dancing
def main():
    # Reset servos to center position
    Arm.Arm_serial_servo_write6(90, 90, 90, 90, 90, 90, 500)
    time.sleep(1)
    times = 0.003
    while True:
        Arm.Arm_serial_servo_write(2, 180-120, time_1)
        time.sleep(times)
        Arm.Arm_serial_servo_write(3, 120, time_1)
        time.sleep(times)
        Arm.Arm_serial_servo_write(4, 60, time_1)
        time.sleep(time_sleep)

        Arm.Arm_serial_servo_write(2, 180-135, time_1)
        time.sleep(times)
        Arm.Arm_serial_servo_write(3, 135, time_1)
        time.sleep(times)
        Arm.Arm_serial_servo_write(4, 45, time_1)
        time.sleep(time_sleep)

        Arm.Arm_serial_servo_write(2, 180-120, time_1)
        time.sleep(times)
        Arm.Arm_serial_servo_write(3, 120, time_1)
        time.sleep(times)
        Arm.Arm_serial_servo_write(4, 60, time_1)
        time.sleep(time_sleep)

        Arm.Arm_serial_servo_write(2, 90, time_1)
        time.sleep(times)
        Arm.Arm_serial_servo_write(3, 90, time_1)
```

```
time.sleep(times)
Arm.Arm_serial_servo_write(4, 90, time_1)
time.sleep(time_sleep)

Arm.Arm_serial_servo_write(2, 180-80, time_1)
time.sleep(times)
Arm.Arm_serial_servo_write(3, 80, time_1)
time.sleep(times)
Arm.Arm_serial_servo_write(4, 80, time_1)
time.sleep(time_sleep)

Arm.Arm_serial_servo_write(2, 180-60, time_1)
time.sleep(times)
Arm.Arm_serial_servo_write(3, 60, time_1)
time.sleep(times)
Arm.Arm_serial_servo_write(4, 60, time_1)
time.sleep(time_sleep)

Arm.Arm_serial_servo_write(2, 180-45, time_1)
time.sleep(times)
Arm.Arm_serial_servo_write(3, 45, time_1)
time.sleep(times)
Arm.Arm_serial_servo_write(4, 45, time_1)
time.sleep(time_sleep)

Arm.Arm_serial_servo_write(2, 90, time_1)
time.sleep(times)
Arm.Arm_serial_servo_write(3, 90, time_1)
time.sleep(times)
Arm.Arm_serial_servo_write(4, 90, time_1)
time.sleep(time_sleep)
time.sleep(time_sleep)

Arm.Arm_serial_servo_write(4, 20, time_1)
time.sleep(times)
Arm.Arm_serial_servo_write(6, 150, time_1)
time.sleep(times)
time.sleep(time_sleep)

Arm.Arm_serial_servo_write(4, 90, time_1)
time.sleep(times)
Arm.Arm_serial_servo_write(6, 90, time_1)
time.sleep(time_sleep)

Arm.Arm_serial_servo_write(4, 20, time_1)
time.sleep(times)
Arm.Arm_serial_servo_write(6, 150, time_1)
time.sleep(time_sleep)

Arm.Arm_serial_servo_write(4, 90, time_1)
time.sleep(times)
Arm.Arm_serial_servo_write(6, 90, time_1)
time.sleep(times)
Arm.Arm_serial_servo_write(1, 0, time_1)
time.sleep(times)
Arm.Arm_serial_servo_write(5, 0, time_1)
time.sleep(time_sleep)
```

```

    Arm.Arm_serial_servo_write(3, 180, time_1)
    time.sleep(times)
    Arm.Arm_serial_servo_write(4, 0, time_1)
    time.sleep(time_sleep)

    Arm.Arm_serial_servo_write(6, 180, time_1)
    time.sleep(time_sleep)

    Arm.Arm_serial_servo_write(6, 0, time_2)
    time.sleep(time_sleep)

    Arm.Arm_serial_servo_write(6, 90, time_2)
    time.sleep(times)
    Arm.Arm_serial_servo_write(1, 90, time_1)
    time.sleep(times)
    Arm.Arm_serial_servo_write(1, 90, time_1)
    time.sleep(times)
    Arm.Arm_serial_servo_write(5, 90, time_1)
    time.sleep(time_sleep)

    Arm.Arm_serial_servo_write(3, 90, time_1)
    time.sleep(times)
    Arm.Arm_serial_servo_write(4, 90, time_1)
    time.sleep(time_sleep)

    print(" END OF LINE! ")

try :
    main()
except KeyboardInterrupt:
    print(" Program closed! ")
    pass

```

```

del Arm # Release Arm object

```